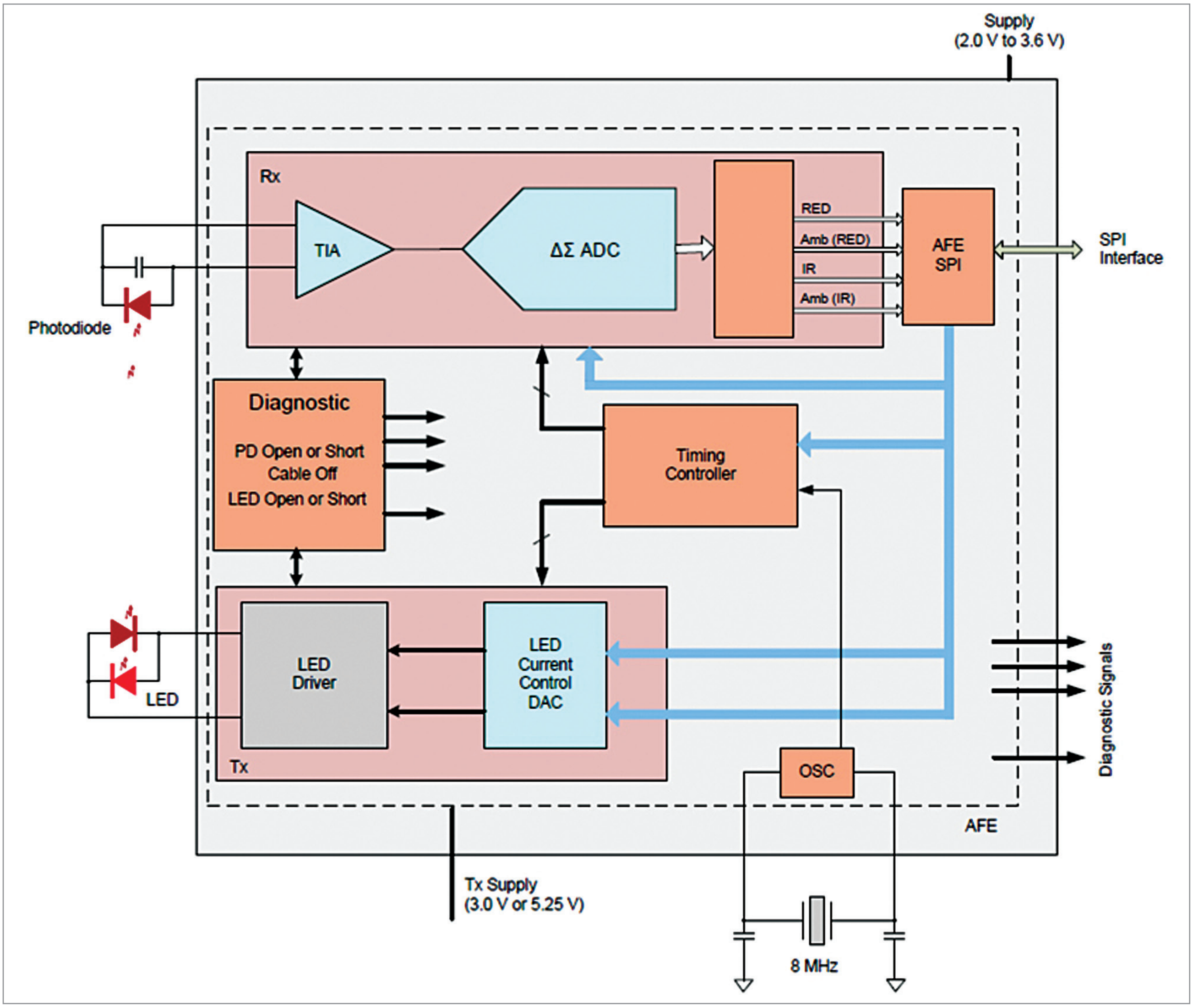

Fig. 3: Block diagram of AFE4403 (Credit: Texas Instruments)

cific wavelengths. This technique is the most common way of heart rate monitoring in wearables.

Main components of an AFE

Given in the Table on previous page are a few circuit blocks that are likely to be present in any AFE. Apart from the other application-specific circuitry present in an AFE, combinations of these blocks are present. It is important to note that for any application involving an AFE, selecting a suitable IC can be a challenge.

Important considerations while selecting AFEs

EMI susceptibility. Like all analogue circuits, AFEs are also susceptible to electromagnetic interferences. Analogue and digital circuits need to co-exist on a single chip, and hence AFEs need good EMI immunity. Since op-amps are sensitive to EMI, it is important to make sure your selected AFE has a good EMI immunity. For example, AFE4403 from Texas Instruments has an EMI filter.

EOS susceptibility. Electrical overstress is a failure state where the device or circuit is subjected to undesirable voltage, current, or power. Analogue circuits are susceptible to EOS as much as they are to EMI. Hence, one must handle AFEs with proper ESD (electrostatic discharge) precautions. Also, check the ESD ratings given in the datasheet.

By understanding the important blocks present in AFEs, the process of selection becomes easier and less time-consuming.

Some analogue front ends

The rest of this article will look at a few AFEs that are particularly tailored for medical applications that involve heart-rate sensing. These are ICs that are not very expensive and can be easily integrated into medical or fitness tech applications.

AFE4403 by Texas Instruments. The AFE4403 is suited for pulse oximeter applications, optical heart rate monitoring as well as industrial photometry applications. This